Submission Mode: Upload `.ipynb` file to Google Classroom

✅ Tasks to Complete:

1. Advanced Scalar & Array Operations

Task:

Perform mathematical operations using a scalar and a custom 1D array.

Requirements:

- Use a scalar value (example: `scalar = 4.3`)
- Create a **1D array containing at least 8 elements** with floating-point values.
- Perform and display results of:

Arithmetic Operations:

- Addition with scalar
- Subtraction with scalar
- Multiplication with scalar
- Division with scalar
- Power operation

Mathematical Functions:

- Square root
- Absolute value
- Exponential function
- Natural logarithm
- Logarithm base 10

Trigonometric Functions:

- sin()
- cos()
- tan()
- arcsin()
- arccos()

- arctan()

Notes:

- Display outputs for all operations.
- Use clipping before applying arcsin() and arccos() to avoid invalid values.

Example:

np.clip(array, -1, 1)

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2. Matrix Mathematical Operations

Task:

Apply mathematical operations on a custom **4×3 matrix**.

Instructions:

- Create a 4×3 matrix with your own floating-point values.
- The matrix should contain both positive and negative values.

Perform and display:

Trigonometric Operations:

- sin()
- cos()
- tan()

Inverse Trigonometric Operations:

- arcsin()
- arccos()
- arctan()

Exponential & Logarithmic Operations:

- exp()

- log()
- log10()

Other Mathematical Operations:

- Absolute value
- Square root
- Square
- Cube root

Rounding and Remainder Operations:

- Remainder when divided by 3
- round()
- floor()
- ceil()

Note:

Handle negative values properly before applying logarithm and square-root functions.

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3. Matrix Statistics and Analysis

Task:

Create a **3×4 matrix** using random floating-point values.

Example:

np.random.uniform(-10,10,(3,4))

Perform and display:

Basic Statistics:

- Maximum value
- Minimum value
- Index position of maximum value
- Index position of minimum value

- Sum of all elements
- Product of all elements
- Mean
- Median
- Variance
- Standard deviation

Row and Column Analysis:

Find:

- Sum of each row
- Sum of each column
- Maximum value from each row
- Minimum value from each column

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4. Matrix Sorting and Searching Operations

Task:

Create a **4×4 integer matrix** using random values.

Example:

np.random.randint(-20,20,(4,4))

Perform the following operations:

Sorting:

- Sort each row in ascending order
- Sort each row in descending order
- Sort the entire matrix in ascending order
- Sort the entire matrix in descending order

Finding Elements:

Display:

- Five largest values in the matrix
- Five smallest values in the matrix
- Positions of elements greater than the matrix mean
- Count of positive values
- Count of negative values
- Count of zero values

Hint:

Use:

np.sort()
np.flatten()
np.where()

✅ Submission Checklist:

Before submission:

✔ All code cells run without errors.
✔ Outputs are clearly visible.
✔ Add comments explaining important code sections.
✔ Handle mathematical errors properly.
✔ Use meaningful variable names.
✔ The notebook file should follow this format:

SIM_CP1_YourName_StudentID.ipynb